

SOP: Troubleshooting Overheated Boiler Feed Pump (Pump P-201)

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Plant/Unit: Sunrise Thermal Power Plant – Unit 3

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Purpose and Scope

This document establishes the standard procedure for troubleshooting an **overheated industrial pump** (Boiler Feed Pump P-201) in the thermal power plant. It outlines the **purpose, safety precautions, step-by-step actions, and diagnostics** to identify and resolve an overheating condition. The scope covers **Pump P-201 (feedwater pump)** in Unit 3 and applies to operations and maintenance personnel. It addresses recognizing overheating symptoms, immediate safety measures, root cause investigation, corrective actions, and safe restart of the pump. This SOP ensures consistent response to pump overheating incidents to **minimize downtime and equipment damage**[\[1\]](#)[\[2\]](#).

Equipment Details

Equipment ID: Pump P-201 (Boiler Feedwater Pump)

Type: Horizontal multistage centrifugal pump (ring-section design) for high-pressure feedwater service.

Service: Feeds condensed water (feedwater) to the boiler economizer at ~180 °C. Designed for continuous operation in a thermal power cycle.

Driver: 3.3 kV Electric Motor (2000 kW) with direct coupling to pump shaft.

Operating Parameters: ~300 m³/hr at 15 MPa discharge (rated), supplying feedwater to maintain drum level. Suction from Deaerator at ~0.5 MPa and 120 °C.

Construction: Eight-stage pump with between-bearing design; journal and thrust bearings are oil-lubricated with an external **oil cooler**. A minimum flow **recirculation line** returns water to the deaerator to prevent low-flow overheating. The pump is installed on a grout-filled baseplate with a flexible coupling to the motor, and is instrumented with temperature sensors on bearings, a vibration monitor, and pressure/flow transmitters.

Symptoms of Overheating

When Pump P-201 is overheating, one or more of the following symptoms may be observed:

- **High Bearing Temperature:** Bearing metal temperature surpasses normal limits (typically normal <70 °C, alarm ~85 °C)[3]. Control room alarms (e.g. “Pump P-201 Bearing Temp High”) may trigger. Prolonged operation beyond **≈ 82–104 °C** can risk seal failure, shaft distortion, and lubricant breakdown[2].
- **Pump Casing Hot to Touch:** The pump casing and bearing housing feel abnormally hot. Infrared thermometer readings show temperature well above ambient. An overheated pump casing (especially if no flow) indicates internal friction or **pump running dry/low-flow** conditions[4][5].
- **Unusual Noise or Vibration:** Overheating bearings may be accompanied by a grinding or howling noise and increased vibration levels[6]. Cavitation (from **vapor lock** or low NPSH) can cause rattling noises and also generates hot spots in the pump[7].
- **Automatic Trip/Shutdown:** The pump may **trip on thermal protection**. Many pumps have built-in safety: e.g. a bearing temperature high-high trip around 95–100 °C or a motor winding temperature trip. If the pump driver is a motor, the motor’s thermal relay might trip due to overheating, stopping the pump.
- **Visible Smoke or Odor:** In extreme cases, overheated oil in bearings can produce a burning smell. Oil may appear dark or **burnt (blackened)** if it has overheated[8]. Paint on the pump or bearing brackets might discolor. Any smoke at the pump or motor is a severe warning sign.

Immediate Safety Actions

Safety is paramount. At the first sign of pump overheating, take these immediate actions to ensure personnel and equipment safety:

1. **Notify and Assess:** Inform the Control Room and shift supervisor. If the pump hasn’t already tripped, evaluate if it’s safe to keep running briefly or if an emergency shutdown is needed. Often, a boiler feed pump has a standby; be prepared to start the **standby pump** and transfer load if possible, to allow P-201 to be taken offline safely.
2. **Reduce Load (If Running):** If the process allows, **slowly reduce the pump’s load or flow** to lessen internal heating[9]. Do not abruptly stop a pump supplying an operating boiler unless the standby is running, to avoid upsetting boiler water supply. If the pump is not critically needed (e.g. unit can be tripped or backed down), commence a controlled shutdown.
3. **Emergency Shutdown:** If the pump shows signs of imminent failure (rapidly rising temperature, smoke, or vibration), **trip the pump immediately**. Use the local **Emergency Stop** or instruct the control room to stop the motor. Verify the pump has stopped rotation. **Close the discharge valve** quickly after trip (if not automatic) to prevent backflow, and crack open the pump vent or a top vent valve to avoid vacuum collapse as it cools.

4. **Lockout/Tagout (LOTO):** Apply full energy isolation. **Electrically isolate the pump driver** by opening the motor breaker and applying a padlock and tag on it (per LOTO procedure). **Isolate fluid energy** by closing the suction and discharge isolation valves. Attach locks and warning tags on valves and switches, indicating “Equipment Under Maintenance – Do Not Operate”^[10]. Relieve any trapped pressure safely: open drain valves to depressurize the pump casing into a suitable container or drain (caution: hot water!). Verify zero energy state – no pressure, no flow, and motor cannot start^[11].
5. **Personal Protective Equipment:** Don appropriate PPE before approaching the pump^[12]. At minimum: **insulated gloves** (to handle hot surfaces), **face shield and safety goggles** (hot water/steam splash risk), **arc-rated coveralls** (if electrical hazards), **hard hat**, and **steel-toe boots**. Ensure no skin is exposed to potential high-temperature spray.
6. **Area Safety and Cool-Down:** Barricade the area if necessary to keep personnel at a safe distance. Post signage like “HOT SURFACE” on the pump. Allow the pump to **cool naturally**. Do not touch or open it until temperature falls to a safe level (typically below 40 °C on the surface). Use fans or ventilation to dissipate heat if in an enclosed space, but **do not apply cold water** on hot casings (risk of cracking).
7. **Fire Watch:** If any smoke or fire was observed, have a fire extinguisher on hand and a fire watch in place. Overheated oil or insulation could ignite. Confirm no fire risk before proceeding.
8. **Hazard Assessment:** Before troubleshooting, perform a hazard assessment of the area^[13]. Ensure no slip hazards (e.g. oil that may have leaked – clean it up), verify adequate lighting, and confirm that any adjacent equipment is stable.

Do NOT proceed with further investigation or disassembly until the pump is fully isolated, locked out, depressurized, and cooled. Once these immediate safety measures are in place, troubleshooting can begin.



Before troubleshooting, ensure the pump is fully isolated and all safety protocols are followed. Lock out the power and valves, and wear appropriate PPE while the equipment cools. Trained technicians must verify zero energy (electrical and pressure) prior to hands-on inspection[10][12].

Step-by-Step Troubleshooting Procedure

Follow this systematic procedure to identify the cause of the overheating and correct it. All steps should be logged in the work order or maintenance log, including measurements and findings at each stage.

1. **Verify Overheat Condition:** Confirm that the pump indeed experienced overheating and it's not a false alarm. Check the bearing temperature gauges or electronic readings for P-201 in the control system. Use an **infrared thermometer** to independently measure the bearing housing and pump casing temperature[14]. Note the highest temperature observed and compare with allowable limits (e.g., >85 °C which is above normal alarm)[2]. Also verify if any **sensor malfunction** could be a factor (e.g., a faulty temperature sensor giving false reading[15]). If possible, inspect the temperature element wiring and calibration dates.
2. **Review Operating Status:** Determine how the pump was operating at the time of overheating. Gather data from the DCS or pump log: flow rate, discharge pressure, suction pressure, motor amps, and any recent transients. An **overheating event** can be exacerbated by operation outside normal range – for example, running the pump at **very low flow (dead-heading)** or **below minimum flow** can cause the pump to overheat due to internal fluid friction[5]. Conversely, running at excessive flow (overload) can overheat the motor and bearings[16]. Note if the pump was in recirculation mode, or if the minimum flow recirc valve was stuck closed. If the

feedwater tank (deaerator) level or pressure was abnormal (causing suction issues), record that as well.

3. **Switch to Standby (if available):** If not already done, start the standby feed pump (P-202 if installed) to take over feedwater supply. Transfer the boiler feedwater load gradually to the standby pump. Once the standby is up to speed and carrying load, **shut down the overheated pump** (if it hasn't tripped already). This ensures you can troubleshoot without impacting unit operation. **Note:** Always adhere to changeover procedures – open the discharge of standby pump, close the discharge of P-201 after current equalization, etc., to avoid water hammer or loss of feed.
4. **Isolation and Drain (re-check):** Re-verify that Pump P-201 is electrically and hydraulically isolated per the **Immediate Safety Actions** section. Double-check that tags are in place and that there is no residual pressure. Open the pump casing vent valve at the top to ensure no pressure builds up due to residual heat. Open a drain valve at the bottom of the pump casing to collect a sample of fluid (caution: it may still be hot water). This fluid sample can be checked for signs of overheating (e.g., if water flashed to steam or any oil presence due to seal leak). Then close drains/vents once pressure is relieved.
5. **External Visual Inspection:** Perform a thorough external check of the pump assembly:
6. **Lubrication Oil Level:** Look at the sight glass on the bearing oil reservoir. The oil level should be at the marked midpoint. If it's low or empty, that's a major cause of overheating (inadequate lubrication)[17]. Top up the oil to proper level *only after cooling*, and investigate any oil leakage that caused the loss[17].
7. **Lubrication Oil Condition:** Observe oil color through the sight glass or dipstick. Dark, blackened, or cloudy oil indicates thermal breakdown or contamination. Smell the oil – a burnt odor means it has likely overheated and lost lubricity[8]. Plan to **drain and replace** degraded oil after internal inspection.
8. **Cooling Water Lines:** Check the **bearing oil cooler** or cooling jacket lines. Ensure the cooling water supply valve is open and water is flowing. Carefully touch (or use IR) on the inlet vs. outlet piping of the oil cooler – a significant temperature difference should be felt if cooling is working. Little to no difference could mean a **blocked cooler or no flow**[18]. Look for any closed valves or clogged strainers in the cooling circuit and open/clean them as needed[18].
9. **Pump and Motor Coupling:** Inspect the coupling for signs of misalignment or damage. Look for **rubber element debris** (if a flexible coupling), cracked coupling bolts, or shifted coupling hubs. Any misalignment can cause **excessive bearing loading and heat**[19]. If the coupling guard is removed, check alignment with a straightedge initially; final alignment will be measured later.
10. **General Area:** Check for any oil sprayed around (could indicate a high-pressure oil leak from a seal or gasket). Notice any **discoloration** on the pump casing or motor (bluing of metal can indicate overheating). Also verify if the pump's **mechanical seal** is intact; overheating can cause seal failure and leaks. Any visible water or oil leak at the seal drain should be noted.

11. **Check Bearing Lubrication System:** If Pump P-201 has an external lube oil system (oil pump, filters, cooler), inspect it in detail:
12. **Oil Pump & Pressure:** Ensure the lube oil pump (if applicable, sometimes shaft-driven or electric auxiliary pump) is operational. Check the lube oil pressure gauge; a low reading might indicate the pump failed or a clogged filter. Normal lube oil pressure should be as per OEM spec (e.g., ~2–3 bar). If zero, that explains the overheating (no oil flow).
13. **Oil Filters:** Isolate and open the oil filter housing (after ensuring no pressure) to see if it's clogged. A clogged filter can starve bearings of oil. Replace or clean filters as needed.
14. **Oil Cooler:** If not already, open any bypass valves and flush the oil cooler. Remove cooler end-caps (if design allows) to inspect for scaling or blockage. Clean the cooler tubes. Restore cooling water flow and verify no blockage.
15. After any interventions, **refill oil** to the correct level with the proper grade (use OEM-specified oil, e.g. ISO VG46 turbine oil). Do not reuse old overheated oil.
16. **Detailed Bearing Inspection:** Once the pump is sufficiently cool, consider inspecting the bearings themselves:
17. Remove the bearing housing covers (journal and thrust bearing housings). This may require unbolting and carefully lifting the covers. **Examine the bearing surfaces** for discoloration (a dark brown or black streak indicates overheating), wiping, or metal smear. Overheated babbitt bearings may show **babbitt melting or wiping**. Rolling-element bearings may show blueing of races or rollers if overheated. If such damage is present, the bearings will need replacement.
18. Check bearing clearances (with a feeler gauge or by shaft lift test) if possible, to see if they are within spec. Overheating can either cause clearance to tighten (thermal expansion) or excessive wear (increasing clearance).
19. Collect an **oil sample** (if not already) and consider sending it for analysis – high metal content or varnish would confirm internal wear and overheating[20].
20. **Shaft and Rotor:** Try to rotate the pump shaft by hand (with coupling disconnected, using a turning wrench). It should turn smoothly. Any binding or rubbing sensation could indicate internal contact (e.g., between impeller and casing or wear rings). If internal rubbing is suspected due to thermal expansion, note that as a possible cause[21].
21. **Internal Pump Inspection (if required):** If external and bearing inspections have not identified a clear cause, you may need to open the pump casing for internal inspection (this is a more involved process requiring pump outage and new gaskets/seals):
22. Drain the pump completely. Unbolt the appropriate casing sections as per the pump's maintenance manual (multistage pumps often have through-bolts and casing split rings).
23. Inspect impellers and diffusers for signs of **cavitation** (pitted surfaces) or **rubbing** (shiny contact marks where clearance should exist). **Cavitation** is a common cause

of overheating and damage – it occurs if NPSH available is too low, causing vapor bubbles that collapse and generate heat[22][23]. If you find cavitation damage, likely the pump was starved of NPSH or run at too high temperature on suction.

24. Check wear ring clearances and balance drum (if any) for tight spots or scoring. Thermal distortion can cause internal parts to rub.
25. Look for any foreign object or debris in the pump that could have partially blocked flow (though unlikely in feedwater line with strainers, but worth a check). A partially blocked suction strainer can cause both cavitation and reduced flow, leading to overheating[23].
26. After inspection, if damage is found (e.g., impeller wear, seized bearing, etc.), plan for repair or replacement of those parts. Document all findings with photos if possible.
27. **Check Alignment & Drive Train:** Using precision tools, measure the pump-motor alignment. Employ a dial indicator or laser alignment tool on the coupling. Check both offset and angular alignment. Misalignment beyond tolerance (typically >0.05 mm for high-speed machinery) can cause bearing stress and overheating[19]. If misaligned, schedule a proper alignment of the motor to pump before restart. Also inspect the baseplate, grouting, and hold-down bolts – a loose foundation can lead to vibration and misalignment over time. Tighten any loose bolts (coupling or base).
28. **Review System Conditions:** Concurrently, evaluate if **systemic issues** contributed: Was the feedwater temperature unusually high (e.g., deaerator malfunction leading to higher-than-design suction temp)? High suction temperature can reduce cooling margin and contribute to pump overheating[24]. Ensure the deaerator is operating at the correct pressure and temperature. Check if the pump was operating continuously without rotation – continuous operation without regular swap can accumulate heat (pumps are often alternated to allow cooling)[25]. If no rotation schedule was in place, this might be a contributing factor.
29. **Determine Root Cause:** Summarize the findings from steps 5–10. Likely root causes of the overheating should become clear. Common findings might include: *low oil level, oil cooler blockage, failed bearing, misalignment, cavitation due to suction issues, or running the pump outside its recommended range*. It's possible multiple factors coincided. Identify all that apply. For each root cause identified, plan the corrective action (see **Root Cause Diagnostics** below for typical causes/remedies). If any cause is still not clear, consider consulting the pump OEM or a specialist, especially if internal damage is puzzling.
30. **Corrective Actions:** Implement repairs and corrections based on the root cause:
 - Replace damaged bearings with new ones (or an entire bearing cartridge if applicable). Ensure new bearings are installed per OEM procedure, with proper clearances and lubrication.
 - Fix the lubrication issues: repair leaks, refill with correct oil, replace filters, and ensure the cooling system is functional (open valves, clean coolers). After refilling oil, **vent the oil system** of air and ensure flow.

- If misalignment was found, re-align the pump and motor. Use laser alignment tools for precision and secure all mounting bolts afterward.
- For cavitation issues, address the upstream cause: ensure suction strainers are clean, the dearator pressure/level are stable, and check that any minimum flow recirculation valve is operating so the pump never runs at <30% design flow[5]. If the recirculation system was faulty (e.g., a closed minimum flow valve), repair or adjust it.
- Replace any worn internal parts (impellers, wear rings, mechanical seal, etc.) as needed before reassembly.
- **Documentation:** Document all parts replaced and adjustments made. If the pump required major overhaul, note it in the equipment history. Also document the cause for future reference (e.g., “Oil cooler 70% blocked with scale – cleaned” or “Found bearing wiped due to oil starvation – replaced and fixed leak at seal”).

31. Pre-Startup Check: Before restarting the pump, perform an inspection to verify all steps are completed: All components are reassembled correctly, oil is at proper level, cooling water is flowing, coupling is reinstalled and aligned, all tools and rags are removed from the area, and **all LOTO tags are still in place** (we will remove them right before restart). Fill out the inspection **Checklist** in the next section to ensure nothing is missed.



During troubleshooting, carefully inspect gauges, oil levels, and cooling lines for anomalies. Systematically check each potential cause of overheating – from lubrication to alignment – and record the findings[26][18]. In the image, a technician examines pump pressure gauges as part of the diagnostic process.

Root Cause Diagnostics

Based on the troubleshooting findings, categorize the **root cause** of the overheating. Common root causes for an overheated pump in a power plant and their diagnostic indicators include:

- **Inadequate Lubrication: Low oil level or poor oil quality** leads to insufficient lubrication, causing bearing metal-to-metal contact and heat[17][27]. Indicators: Low sight-glass reading; burnt oil smell; bearing damage observed. *Remedy:* Restore oil to proper level, fix leaks, use correct oil grade, and replace bearings if damaged. Implement regular oil level checks and scheduled oil changes.
- **Cooling System Failure: Blocked or reduced cooling flow** to the oil cooler or bearing housing causes temperatures to rise[28]. Indicators: Little temperature drop across cooler; fouling in cooling water passages; high oil temperature. *Remedy:* Clean/flush the oil cooler and cooling lines[18], ensure valves are open and water quality is good. Consider installing filters or water treatment if scaling is recurrent.
- **Misalignment or Mechanical Friction: Misaligned pump and driver shafts** cause extra load on bearings and seals, generating heat[19][29]. Internal rubbing (e.g., due to thermal expansion or misassembly) can also overheat the pump[21]. Indicators: Coupling wear patterns, high vibration, uneven bearing wear. *Remedy:* Realign the pump and motor to within tolerance; verify proper thermal expansion procedures (warm up pump if needed, allow sliding foot if applicable). Ensure internal clearances and alignment pins are per OEM spec during reassembly.
- **Cavitation / Suction Issues: Cavitation** occurs if Net Positive Suction Head (NPSH) available is too low or feedwater temperature is too high, causing vapor bubbles that collapse and generate local heat[7][22]. Also, **low flow operation** (pump running with discharge closed or <30% flow) causes continuous recirculation of hot water inside the pump[5]. Both result in overheating of fluid and pump. Indicators: Noise (gravel-like or popping sounds), pitted impeller surfaces, possibly oscillating discharge pressure. *Remedy:* Fix suction conditions – maintain dearator pressure/level, clean suction strainers[23], and ensure minimum flow line is functioning. Avoid operating pump below the minimum flow; if load is low, keep recirculation open or use auto-recirculation valve.
- **Overloading / Operating Beyond Design:** Pushing the pump beyond its design point (excess flow or head) or running at full load continuously can overheat the motor and bearings[16][30]. Indicators: Motor current above rated, motor casing very hot, frequent over-current trips, or discharge pressure/flow higher than spec. *Remedy:* Throttle flow to keep pump within its performance curve. If process demands have changed, consider a pump with proper sizing. Ensure safety devices (like thermal overloads and flow limiters) are set correctly.
- **Bearing Wear or Failure:** A bearing nearing end-of-life (fatigued or contaminated) can run hot even if lubrication is ok, due to increased friction. Indicator: Vibration

trending upward over time, metal in oil sample, a wobbly shaft or unusual shaft movement. *Remedy:* Replace bearings at scheduled intervals; investigate any bearing that fails prematurely for underlying cause (misalignment, contamination, etc.).

- **Continuous Operation without Rest:** Pump P-201 might have overheated simply due to lack of rotation – running non-stop for extended periods can cause gradual heat soak, especially if its standby was never used[25]. Even if within design, continuous running can wear out oil and components. *Remedy:* Implement a **pump rotation schedule** (e.g., alternate pumps weekly or monthly)[25] so each pump can cool down and undergo maintenance. Ensure oil is changed at recommended intervals to avoid degradation.
- **External Environmental Factors:** Less common in a controlled power plant environment, but high ambient temperature or poor ventilation around the pump can impede heat dissipation[31]. For example, if the pump room ventilation failed, heat could accumulate. Also, check for any external heat sources near the pump. *Remedy:* Restore room ventilation fans, remove any heat barriers around pump, and in extreme cases consider adding a fan or cooling unit for the pump area.
- **Instrumentation Issues:** In some cases, the “overheating” might have been a false reading from a bad sensor. If no evidence of actual overheating is found, suspect the instrumentation. *Remedy:* Calibrate or replace faulty temperature sensors, and verify the trip settings.

Each cause above may contribute to the overheating. It’s crucial to **address all contributing factors**, not just one. For instance, if misalignment and low oil were both found, both issues must be resolved. Reference relevant OEM manuals for tolerance and recommended practices for each fix. Document the root cause analysis in the report and update the preventive maintenance plan to avoid recurrence (e.g., schedule cooler cleaning, alignment checks, etc.).

Supporting Info: Overheating in pumps is often multifactorial. Industry guidelines (e.g., NFPA 20 for fire pumps) note that bearing temps beyond ~82–104 °C are unsafe[2]. Main causes identified in industry case studies include lubrication failures, blocked cooling pathways, rotor misalignment, cavitation from suction issues, and overloading conditions[32] – all of which have been considered in the above analysis.

Inspection Checklist – Overheated Pump P-201

After completing repairs or adjustments, use the following **inspection checklist** to ensure all necessary items have been addressed and the pump is ready for a safe restart. Each checkpoint should be reviewed and marked off by the responsible engineer or technician:

Inspection Item	Acceptable Condition / Criteria	Status	Remarks
		(✓/X)	
LOTO Verification	All energy sources isolated		e.g., Breaker

Inspection Item	Acceptable Condition / Criteria	Status (✓/X)	Remarks
	(electrical, hydraulic).		tagged, valves locked.
Pump Casing Cool	Casing temperature < 40 °C (safe to touch).		(Use IR gun to confirm).
Oil Level (Bearing Housing)	Oil at midpoint of sight glass; no visible leaks[17].		(Refilled with ____ oil grade).
Oil Condition	Clean, amber color; no burnt smell or turbidity.		(Oil changed on mm/dd/yy).
Oil Pressure (if applicable)	> 2 bar during priming (when pump running).		(Measured ____ bar at idle).
Cooling Water Flow	Cooling water valves open; cooler inlet vs outlet warm/cool to touch[18].		(Inlet °C / Outlet °C).
Bearing Temperature	Normal range 50–70 °C during operation[3]; Alarm set ~85 °C.		(Verify sensor reading).
Bearing Replacement	Bearings inspected/replaced if needed; clearances OK.		(New DE/NDE bearings installed?).
Coupling Alignment	Alignment within 0.05 mm (shim adjusted as needed)[19].		(Laser alignment report attached).
Coupling Hardware	Coupling bolts tight; elastomer elements in good condition.		(Replaced spider element).
Internal Inspection	Impellers, wear rings, seal in good condition (no rubbing/cavitation damage).		(Impeller #3 replaced? Y/N).
Suction Strainer	Strainer/basket clean; no debris or blockage[23].		(Cleaned on mm/dd/yy).
Mechanical Seal	Seal not leaking; flush/quench lines (if any) flowing.		(Seal replaced? Y/N).
Bolting & Supports	All pump and base bolts torqued; support clamps secure.		(Baseplate grout intact).
Instrumentation	Temp and vibration sensors calibrated; alarm/trip setpoints verified.		(Calibrated on mm/dd/yy).
Standby Equipment	Standby pump in auto or ready; no issues after swap.		(P-202 returned to standby).
Documentation	Work order updated; cause and fix		(Report # filed in

Inspection Item	Acceptable Condition / Criteria	Status (✓/X)	Remarks
	documented; crew sign-off done.		CMMS).

Instructions: For each item, enter a check ✓ if okay or X if not, and add any remarks or corrective actions taken. All items must be ✓ (acceptable) before removing LOTO and restarting Pump P-201.

Restart Procedure After Repair

Once the pump has been inspected and all issues rectified, follow this controlled restart procedure to safely return Pump P-201 to service:

1. **Pre-Start Briefing:** Conduct a brief with Operations and Maintenance personnel. Review the work done and ensure everyone is aware of the restart plan. Confirm that it's safe to restart and all team members are clear of the pump area.
2. **Remove Lockouts/Tagouts:** The responsible person (maintenance lead) should remove the LOTO devices in coordination with the Control Room. Restore power supply to the motor (close breaker) **only when instructed**, and crack open (partially open) the suction and discharge isolation valves. Ensure the discharge valve is initially **mostly closed** – do **not** start the pump on a wide-open discharge valve[33], as this can cause a sudden load and “jerk” the system.
3. **Prime and Vent:** If the pump was drained or opened, ensure it is fully refilled with water (prime the pump). Open the vent valve on top of the pump casing and fill the pump either by gravity from the deaerator or using a manual filling pump until water comes out of the vent without air. Close the vent. Verify suction valve is fully open and suction pressure is normal (deaerator pressure stable).
4. **Cooling & Lube Startup:** Start any auxiliary systems first: turn on the cooling water flow to the bearings/oil cooler and the lube oil pump (if a separate lube pump exists) to establish lubrication before the main pump starts. Verify oil pressure is in range and cooling water is flowing.
5. **Bump Check (Rotation):** Engage the motor briefly (bump start) to verify the rotation direction is correct (especially if motor leads were disconnected during work). The pump should rotate in the correct direction (check arrow on casing). Also listen for any unusual sounds on this bump. Then stop. If rotation is correct and no immediate issues, proceed.
6. **Start Pump:** Start the pump either locally or from the control room per standard operating procedures. The pump will come up to speed (for motor-driven, this is quick; for steam turbine-driven, follow the warm-up and rolling procedure as applicable). **Monitor the pump closely** during startup. Keep the discharge valve ~5–10% open initially to allow a minimum flow out (prevent dead-head) but still impose backpressure so the pump isn't at full load instantly.

7. **Gradually Load Pump:** Once at full speed and stable, slowly open the discharge valve to allow the pump to ramp up to the desired flow/pressure. Do this in coordination with the control room to smoothly transfer feedwater flow from the standby pump (if applicable) or to meet boiler demand. Avoid any rapid flow changes to prevent water hammer.
8. **Monitor Conditions:** As the pump reaches normal operating conditions, **closely monitor all parameters:** bearing temperatures, oil pressure, cooling water temperatures, pump discharge pressure/flow, motor amps, and vibration readings. Initially, check bearing temperature every 5 minutes; it should rise slowly and stabilize within normal range (e.g., if it was 25 °C ambient, it might stabilize around 60 °C)[3]. Vibration should remain below alarm (typically <5 mm/s). Listen for any abnormal noise (no rattling or grinding should be present).
9. **Leak Check:** Inspect all around the pump for leaks as it runs: look at mechanical seal (should be no excessive dripping, slight weep could be normal if it's a packed gland), check all flanges and drains. Tighten any packing gland slightly if needed (if it's packing type seal) to stop a steady drip – but do not over-tighten.
10. **Normal Operation:** When the pump is fully on-line and stable, notify the control room. The pump can now carry the feedwater load. If this pump was a standby now replacing a failed pump, ensure the system is adjusted accordingly. If both pumps will run in parallel for a time, coordinate their outputs to desired levels.
11. **Post-Start Monitoring:** For at least the first hour of operation, have maintenance personnel periodically (every 15 minutes) check the pump's temperature and other vital signs. Ensure the bearing temperature stays well below alarm. Any trending upward could indicate an unresolved issue – be prepared to take action if something isn't right.
12. **Remove Temporary Controls:** Remove any tags or signs that were placed around the pump area indicating it was under maintenance once fully satisfied the pump is running okay. (However, do keep any "Caution Hot" signs if surfaces are still above 50 °C).
13. **Documentation and Handover:** Formally hand over the pump to Operations. Document the restart date/time and the conditions in the maintenance log. Also schedule a follow-up inspection after 24 hours of runtime to re-check bolt tightness, oil level, etc. Finally, close the work order with detailed notes on the cause and fix. If any **Management of Change (MOC)** or engineering follow-up is required (e.g., updating a procedure or design change like adding a bearing temperature sensor), note that for action.

Reminder: Never start or stop the pump with the discharge valve fully open, to avoid water hammer and mechanical stress[33]. Starting against a closed (or slightly open) discharge ensures a smoother ramp-up and prevents sudden jerk loads on the pump and driver, which can damage bearings and internal components[33]. Always follow the OEM start-up guidelines specific to Pump P-201.

Common Causes and Historical Notes

Overheating issues with Pump P-201 have occurred in the past. Understanding these historical cases can help prevent future recurrences. Below are some **common causes of past overheating incidents** and notes from our plant's history:

- **June 2024 Incident:** Pump P-201 tripped on high bearing temperature. Investigation found the **lube oil cooler was partially blocked** with sediment. Cooling water flow was only ~50%. The reduced heat removal caused gradual temperature rise over a shift. The bearing metal temp reached 95 °C triggering a trip. **Action:** The cooler was chemically cleaned and a cooling water filtration system was installed to prevent blockage. After this fix, normal bearing temps dropped to ~60 °C under full load. This incident reinforced the importance of regular cooler maintenance.
- **December 2024 Incident:** After an overhaul, P-201 experienced overheating and high vibration on restart. It did not trip, but temperatures were ~10 °C above normal and rising. We shut it down as a precaution. Root cause was **misalignment** – the pump and motor were not precisely aligned during re-installation. This caused excessive coupling strain and hot bearings. **Action:** Performed laser alignment (found angular misalignment of 0.3 mm at coupling). Post-alignment, the vibration and temperature returned to normal. Lesson learned: always perform and verify alignment after maintenance.
- **Continuous Operation Issue (General):** It was noted in 2023 that Pump P-201 ran for 14 months continuously without rotation, because the standby pump P-202 was out of service for overhaul. The extended run led to gradual oil degradation and a minor increase in bearing temperature over time. While no trip occurred, the bearing temps were ~5 °C higher than when pumps are swapped regularly. **Action:** We now adhere to a **pump changeover schedule** – swapping P-201 and P-202 monthly[25]. This allows each pump to cool, and lets maintenance check oil and components. Since implementing rotation, bearing temperatures have remained consistently within range.
- **OEM Advisory (2025):** The pump manufacturer (ABC Pumps Co.) issued a tech bulletin advising to check the minimum flow recirculation orifice for clogging. A partially clogged recirc line can reduce bypass flow at low loads, potentially causing pump overheating due to low-flow operation. We inspected our recirculation line and found some scaling. **Action:** In the next outage, we plan to clean or replace the minimum flow orifice and add it to the maintenance checklist.
- **Common Causes Summary:** Overall, our experience matches industry guidance: most overheating cases trace back to **cooling/lubrication issues, flow extremes, or mechanical setup problems**[32][34]. By addressing oil quality, keeping cooling water clean, operating within design parameters, and ensuring proper installation, we have significantly reduced overheating occurrences.

These historical notes should be shared with new engineers and included in training. They highlight why adhering to this SOP and performing regular preventative maintenance (oil changes, cooler flushes, alignment checks, etc.) is critical.

Safety and Compliance Reminders

Working on high-pressure, high-temperature feed pumps involves significant hazards. Always keep these safety and compliance points in mind:

- **Lockout/Tagout:** Never bypass or shortcut the LOTO process. Before maintenance, all energy sources **must** be isolated and locked/tagged[10]. Verify isolation each time (try to start the motor with the breaker open to confirm zero energy). This is not only plant policy but also an OSHA requirement for hazardous energy control.
- **Personal Safety:** Wear appropriate PPE at all times and be aware of hot surfaces. Use insulated tools/gloves on parts that might be hot. If you must work near a running pump (for inspections), wear hearing protection and stay clear of rotating parts (ensure guards are in place). Do not wear loose clothing or jewelry that could get caught.
- **Pressure Hazard:** Treat any pressurized section with caution. Even after shutdown, wait and open drains slowly, as hot water under pressure can cause *violent flash steam*. Stand to the side of valves/flanges when cracking them open. Ensure relief paths are working (like thermal relief valves on isolated sections).
- **Environmental and Regulatory Compliance:** Dispose of any waste oil, oily rags, used filters, and damaged parts according to environmental regulations. Document the disposal per the plant's waste management plan. Any release of feedwater or oil should be reported and cleaned up to prevent slips or environmental harm.
- **Follow OEM Guidelines:** This SOP complements the OEM manual – always cross-reference the pump manufacturer's recommendations for limits and procedures. For example, OEM may specify maximum casing heat-up rates, specific lubrication grades, or unique restart steps. Ensure compliance with those to maintain warranty and integrity.
- **Recordkeeping:** It's crucial to log all maintenance activities in the Computerized Maintenance Management System (CMMS) or logbook. Accurate records support reliability analysis and compliance audits. Note the date, findings, parts replaced, clearances measured, and persons involved. Also log the bearing temperature trends and vibration readings after the event for future reference.
- **Training and Authorization:** Only qualified personnel should perform this troubleshooting. Electrical work (isolating motor, etc.) must be done by authorized electricians, and mechanical work by trained millwrights or engineers. Ensure everyone has read this SOP and related documents. Conduct drills or reviews of pump trip response periodically.
- **Compliance Standards:** Adhere to relevant standards such as OSHA 1910 for general industry safety, and any power industry regulations. For example, NFPA

standards may not directly apply to feed pumps (they apply to fire pumps), but their guidelines on temperature limits can be a helpful reference[2]. Company engineering standards for rotating equipment should be followed. If an incident occurs, it may need to be reported as per regulatory requirements (e.g., if there was a reportable release or if safety interlocks failed). Always aim for not just compliance but a safety-first culture.

By keeping safety at the forefront and following this procedure, maintenance staff can troubleshoot and restore the pump without incident, and ensure reliable and safe operation in the power plant.

Appendix

A. Instrumentation Alarm/Trip Setpoints (Pump P-201)

(For reference – actual setpoints as per Instrumentation Dept.)

Parameter	High Alarm	High-High Trip	Notes
Pump Bearing Metal Temperature	85 °C (alarm)	95 °C (trip)	Normal ~50–70 °C[3]. Trip based on OEM limit ~100 °C[2].
Lube Oil Temperature (outlet)	60 °C	70 °C	At oil cooler outlet. High temp may indicate cooling issues.
Lube Oil Pressure	1.5 bar (low alarm)	1.0 bar (low trip)	Measured at bearing header. Trips pump on loss of oil pressure.
Pump Casing Drain Temperature	120 °C (alarm)	– (no trip)	(If > alarm, indicates possible flash or low flow heating).
Pump Vibration (Overall)	8 mm/s RMS (alarm)	12 mm/s RMS (trip)	Per ISO 10816 for this class of machine.
Motor Winding Temperature	130 °C (alarm)	140 °C (trip)	Motor thermal limit (if motor-driven).
Motor Bearing Temperature	90 °C (alarm)	100 °C (trip)	Motor has its own bearings; similar limits as pump.
Suction Pressure	0.3 MPa (low alarm)	–	Low suction can cause cavitation alarm. (No trip, but an interlock may prevent start if too low).
Discharge Pressure	14 MPa (low alarm)	16.5 MPa (high alarm)	Alarm if outside normal range (indicative of off-curves operation).

Note: These setpoints are configured in the plant DCS for Pump P-201. **Alarm** alerts operators to take action, **Trip** will automatically shut down the pump to protect it. Values

are based on manufacturer's recommendations and industry practices (bearing temps per OEM, aligned with NFPA 20 guidance of ~82–104 °C max[2]). Always refer to the latest instrument datasheets for any updates to these values.

B. Contact Information

For further assistance or to report an incident involving Pump P-201, use the following contact list:

- **Maintenance Lead – Mechanical:** *John Doe* – Phone: +971-55-123-4567, Email: john.doe@sunrisepower.ae
- **Operations Shift Supervisor (Unit 3):** *Maryam Al Mansoori* – Radio: Channel 3, Phone: +971-55-234-5678
- **Instrumentation Engineer:** *Ahmed Rahim* – Ext: 4431, Email: ahmed.rahim@sunrisepower.ae
- **Maintenance Manager:** *Li Wei* – Ext: 4400, Office: Bldg 5 Room 12
- **Safety Officer:** *Priya Patel* – Ext: 4015, to be notified in case of any safety incident or if permit conditions need review.
- **Pump OEM Support (ABC Pumps Co.):** 24/7 Hotline: +1-800-ABC-PUMP (have pump serial number ready when calling). Local Rep: *Gopal Srinivasan* – Email: gopal.srinivasan@abc-co.com.

Keep this contact list posted in the control room and maintenance workshop. In an emergency (e.g., fire, severe injury), always call the main emergency number (Dial 999 from plant phone or as per site protocol) **before** contacting individuals.

Revision History:

- Rev. 1.0 – 15/12/2025: Initial issue of troubleshooting guide for overheated pump P-201 (Author: J. Smith). Approved by Plant Manager A. Khan.
- Rev. 0.0 – 01/12/2025: Draft created and reviewed by Maintenance and Operations (not released).

References & Related Documents:

- Pump P-201 OEM Manual – ABC Pumps Document #BF-P201-O&M, Rev. 2019.
- Plant Maintenance Procedure ENG-MECH-012: “Rotating Equipment Alignment Standard”.
- HSE Procedure HSE-LOTO-001: “Lockout/Tagout – Control of Hazardous Energy”.
- Incident Report IR-2024-06: “Investigation of P-201 Overheating on 01-June-2024”.
- [35]91FirePump – “How to Handle Fire Pump Overheating Problems”, Sec.1, 2025 – for general temperature guidelines and causes.
- [36][37]PowerPlantCalculations – “Boiler Feed Pump Troubleshooting Guide”, 2025 – on high bearing temperature causes (low oil, cooling).
- [34][5]PowerPlantCalculations – “BFP Pump Overheating Causes”, 2025 – guidance on

operation at low flow and high flow affecting pump temperature.

- [38]EmpoweringPumps – “*Bearings Run Hot – Troubleshooting*”, 2020 – on misalignment and lubrication checks for hot bearings.

- [10][12]MPO Cincinnati – “*Industrial Pump Maintenance Checklist*”, 2025 – Pre-maintenance safety (LOTO, PPE).

(End of SOP)

[1] [4] [6] [16] [31] Troubleshooting Guide: How to Prevent A Pump Overheating?

<https://www.streampumps.com/product-qa/pump-overheating-240604.html>

[2] [7] [8] [9] [14] [20] [22] [27] [28] [29] [30] [32] [35] How to Handle Fire Pump Overheating Problems

<https://www.91firepump.com/media/industry-news/how-to-handle-fire-pump-overheating-problems.html>

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<https://www.mpofcinci.com/blog/industrial-pump-maintenance-checklist/>

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